AI FORGE EXPO

# EduBridge AI

Finding learning gaps early. Bridging them with personalized support.

**SDG 4: Quality Education**

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REAL PROBLEM

# Problem Statement

## The real-world problem

Students move to the next grade without mastering foundational concepts. Early gaps in reading, numeracy and core ideas compound over time, and no system detects them in time.

## Who is affected?

School and college students, especially in under-resourced and large-class settings. Teachers are affected too: they cannot track every learner individually.

## Where and when does it occur?

In classrooms where one pace fits all, from primary years onward. Gaps stay hidden until exams or higher-level topics expose them, often too late.

## Why is it important?

Weak foundations lead to disengagement, dropouts and lost potential. Timely, personalized support turns "falling behind" into "catching up".

EVIDENCE

# Problem Evidence / Research

**~70%**

of 10-year-olds in low- and middle-income countries cannot read and understand a simple text

*World Bank, Learning Poverty*

**~250M**

children and youth worldwide are out of school, and many in school are not learning

*UNESCO, 2023*

**~1 in 4**

Indian youth aged 14-18 cannot fluently read a Std II-level text in their own language

*ASER 2023*

**Research finding:** Bloom (1984) found one-to-one mastery tutoring outperformed classroom teaching by about two standard deviations. Personalization works, but it does not scale without technology.

Sources: World Bank Learning Poverty update; UNESCO GEM Report / UIS 2023; ASER Centre 2023; Bloom, B. (1984). Please verify latest figures before final submission.

SUSTAINABLE DEVELOPMENT GOAL

# SDG Alignment

**4**

## QUALITY EDUCATION

Ensure inclusive and equitable quality education and promote lifelong learning for all

### Target 4.1

Ensure all learners complete quality primary and secondary education with relevant learning outcomes.

### Target 4.5

Eliminate disparities in education and ensure equal access for vulnerable and under-resourced learners.

### How the problem connects

Learners pass through grades without achieving real outcomes, which is the exact gap 4.1 targets. EduBridge AI diagnoses gaps and offers free, adaptive support to close them.

EXISTING GAP

# Existing Solutions

### Current methods

Periodic exams and report cards. Private tuition and coaching. Generic e-learning platforms and video lectures. Teacher observation in class.

### Their limitations

Exams show scores, not the root concept missed. Tuition is costly and unequal. Content is one-size-fits-all. Teachers cannot individually track 40+ students.

### The gap

No affordable, continuous system that pinpoints each student's specific knowledge gaps, recommends the right resource, and tracks improvement over time.

SOLUTION

# Proposed Solution

**EduBridge AI diagnoses what each student does not yet understand, recommends targeted resources, and tracks progress, so no learner is left behind.**

### Gap detection

Adaptive diagnostic quizzes map mastery by concept, not just by score.

### Root-cause analysis

Traces wrong answers back to missing prerequisite topics.

### Targeted resources

Recommends videos, notes and practice matched to each gap and level.

### Progress tracking

Mastery dashboards show improvement over time for students.

### Teacher insights

Class-level heatmaps reveal common gaps and who needs help.

### Accessible by design

Multilingual, low-bandwidth friendly, works on a basic phone.

WORKFLOW

# System Workflow / Architecture

**1**

### Input

Quiz answers, response time, attempts, class and syllabus data

>

**2**

### Processing

Clean data, map each answer to concepts in a topic graph

>

**3**

### AI / Software

Knowledge tracing model estimates mastery; recommender ranks resources

>

**4**

### Output

Gap report, personalized study path, progress score

>

**5**

### User interaction

Student learns and retakes checks; teacher views insights

*Feedback loop: new results update each student's mastery model, and recommendations adapt continuously.*

TECHNOLOGY

# Technology Stack

### AI / ML

Python, scikit-learn, PyTorch; Bayesian / Deep Knowledge Tracing; embeddings for resource matching

### Frontend

React with Tailwind CSS; responsive, mobile-first progressive web app

### Backend

FastAPI (Python) REST services for quizzes, analytics and recommendations

### Database

PostgreSQL for users and progress; vector store for content search

### APIs / Cloud

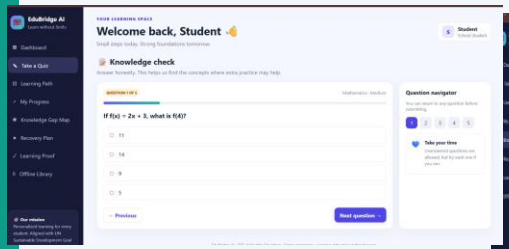
LLM API for explanations; YouTube / open educational resource APIs; cloud hosting (AWS / GCP)

### Hardware

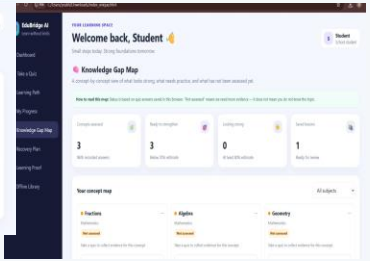
None required. Runs on any smartphone, tablet or school computer.

PROTOTYPE

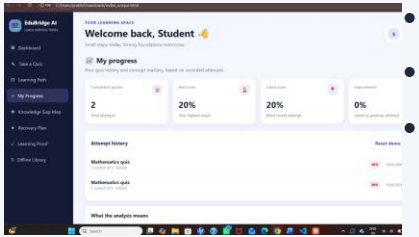
# Prototype / Working Demo




Diagnostic quiz




Gap report




Progress dashboard

## Main features

- Adaptive diagnostic quiz
- Concept-level gap report
- Personalized resource list
- Mastery progress tracker

https://edubridge26.netlify.app/

# Innovation / USP

## What is unique?

Concept-level diagnosis that traces a student's mistake back to the missing prerequisite, then closes the loop by tracking improvement.

## Why is it better?

Continuous and personalized rather than one-off exams. Free resources keep cost near zero. Teachers get clear, actionable insights.

## How is it different?

Existing platforms deliver content or scores. EduBridge AI connects detection, recommendation and tracking in one adaptive, low-bandwidth, multilingual system.

# Impact

**Days, not months**

Faster gap identification vs. exam-based detection

**+20-30%**

Measurable gain in post-intervention mastery scores

**50%+**

Higher student engagement and teacher time saved

## Who benefits?

Students get personalized help. Teachers get class insights. Schools and parents get transparent progress data.

## Contribution to SDG 4

Directly advances Target 4.1 (learning outcomes) and 4.5 (equity) by giving every learner individual support.

*Impact figures are pilot targets to be validated through testing.*

# Feasibility & Scalability

### Cost

Low: open-source stack, free-tier cloud and open educational resources.

### Deployment

Web app / PWA deployed to the cloud; pilot in one school, then expand.

### Practicality

Works on existing phones and school computers with minimal teacher training.

### Scalability

Stateless APIs and cloud autoscaling; new subjects added as topic graphs.

### Challenges

Limited quality data at start, connectivity gaps and content alignment. Mitigate with pilot data, offline mode and curriculum mapping.

### Ethics & privacy

Minimal data collection, consent, anonymized analytics and bias checks.

# Future Scope & References

## Future improvements

Voice-based AI tutor. Offline mobile app. Emotion and engagement detection. Parent and teacher alerts.

## Expansion

More subjects and boards, regional languages, college and skill-training programs, NGO and government school partnerships.

## References


- World Bank: Learning Poverty indicator updates
- UNESCO Institute for Statistics / GEM Report (2023)
- ASER Centre: Annual Status of Education Report 2023
- UN SDG 4 targets: sdgs.un.org/goals/goal4
- Bloom, B. S. (1984). The 2 Sigma Problem
- Piech et al. (2015). Deep Knowledge Tracing